

$$1. A = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \quad \vec{b} = \begin{pmatrix} 8 \\ 12 \end{pmatrix} \quad \vec{c} = (1 \ 3 \ 0 \ 0)$$

$$2. \vec{A}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \vec{A}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \vec{A}_3 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad \vec{A}_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$3. B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad N = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$$

$$4. \vec{c}_B = (1 \ 0) \quad \vec{c}_N = (3 \ 0)$$

$$5. \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \vec{x}_B = B^{-1} \vec{b} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 12 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}, \text{不是最优解}$$

$$6. \Delta \vec{x}_B = -x_2 \cdot B^{-1} \vec{A}_2 = -x_2 \cdot \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = -x_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -x_2 \\ -x_2 \end{pmatrix}$$

$$7. Z = \vec{c}_B^T B^{-1} \vec{b} + x_2 (c_2 - \vec{c}_B^T B^{-1} \vec{A}_2) = (1 \ 0) \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 12 \end{pmatrix} + x_2 (3 - (1 \ 0) \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}) = 8 + 2x_2$$

$\therefore$ 随 x_2 的增大, Z 的值增大

$$8. \Delta \vec{x}_B = -B^{-1} N \vec{x}_N = -\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -x_3 \\ -x_3 \end{pmatrix}$$

$$9. Z = \vec{c}_B^T B^{-1} \vec{b} + (\vec{c}^T - \vec{c}_B^T B^{-1} A) \vec{x} = (1 \ 0) \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 12 \end{pmatrix} + ((1 \ 3 \ 0 \ 0) - (1 \ 0) \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 1 \end{pmatrix}) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 8 + 2x_2 + x_3$$

$\therefore$ 随 x_3 的增大, Z 的值增大

10. 对于基变量检验数的值为0, 非基变量检验数全为负号时当前基解为最优解

$$11. B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \quad N = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \vec{c}_B = (1 \ 3) \quad \vec{c}_N = (0 \ 0) \quad \vec{x}_B = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = B^{-1} \vec{b} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}, \text{是最优解}$$

$$\Delta \vec{x}_B = -x_3 B^{-1} \vec{A}_3 = -x_3 \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = -x_3 \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2x_3 \\ -x_3 \end{pmatrix}$$

$$Z = \vec{c}_B^T B^{-1} \vec{b} + x_3 (c_3 - \vec{c}_B^T B^{-1} \vec{A}_3) = (1 \ 3) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 12 \end{pmatrix} + x_3 (0 - (1 \ 3) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix}) = 16 - x_3$$

$\therefore$ 随 x_3 的增大, Z 的值减小

$$\Delta \vec{x}_B = -B^{-1} N \vec{x}_N = -\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -2x_3 + x_4 \\ -x_3 - x_4 \end{pmatrix}$$

$$Z = \vec{c}_B^T B^{-1} \vec{b} + (\vec{c}^T - \vec{c}_B^T B^{-1} A) \vec{x} = (1 \ 3) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 12 \end{pmatrix} + ((1 \ 3 \ 0 \ 0) - (1 \ 3) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 1 \end{pmatrix}) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 16 - x_3 - 2x_4$$

$\therefore$ 随 x_4 增大, Z 的值减小

对于基变量检验数的值为0, 非基变量检验数全为负号时当前基解为最优解